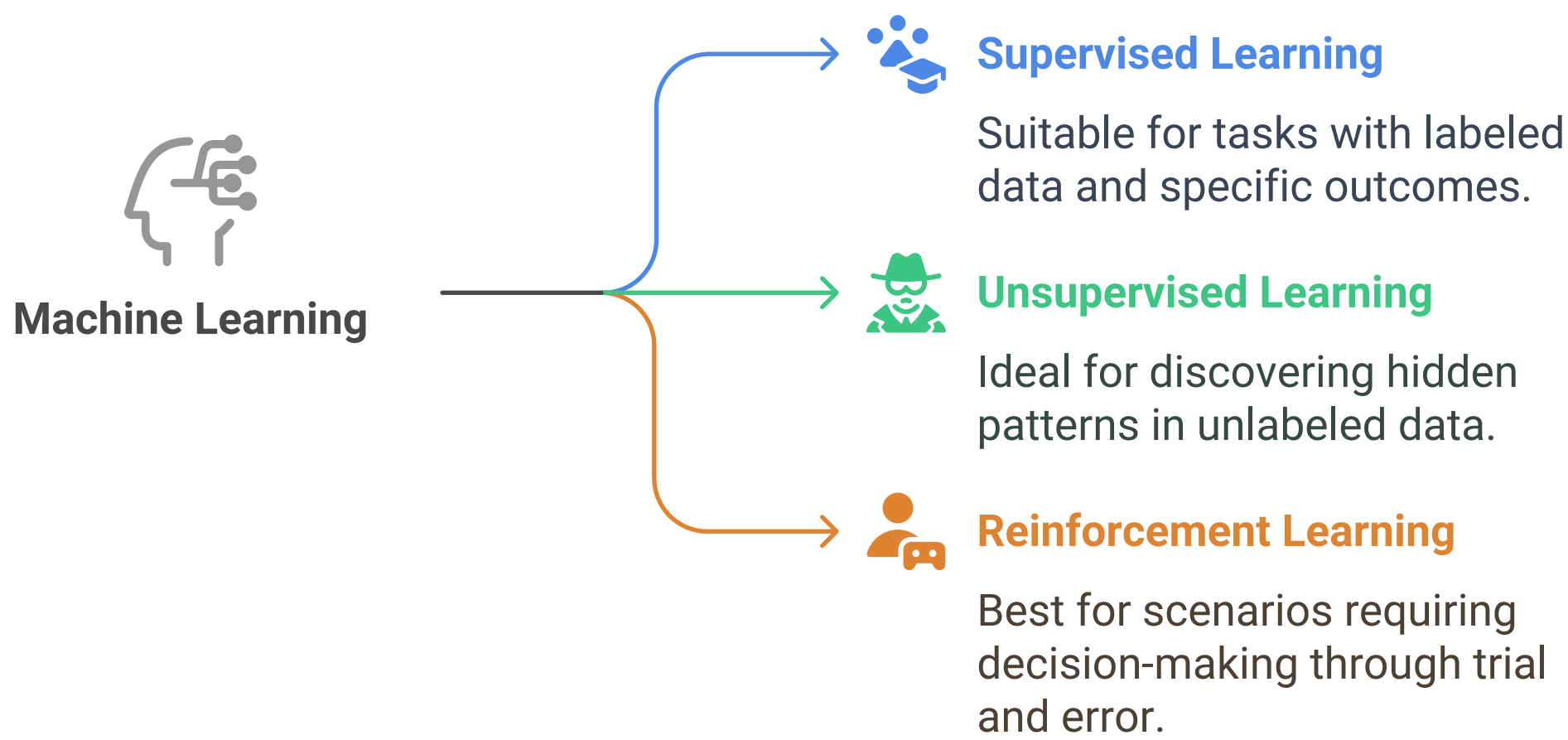


Understanding the Three Types of Machine Learning

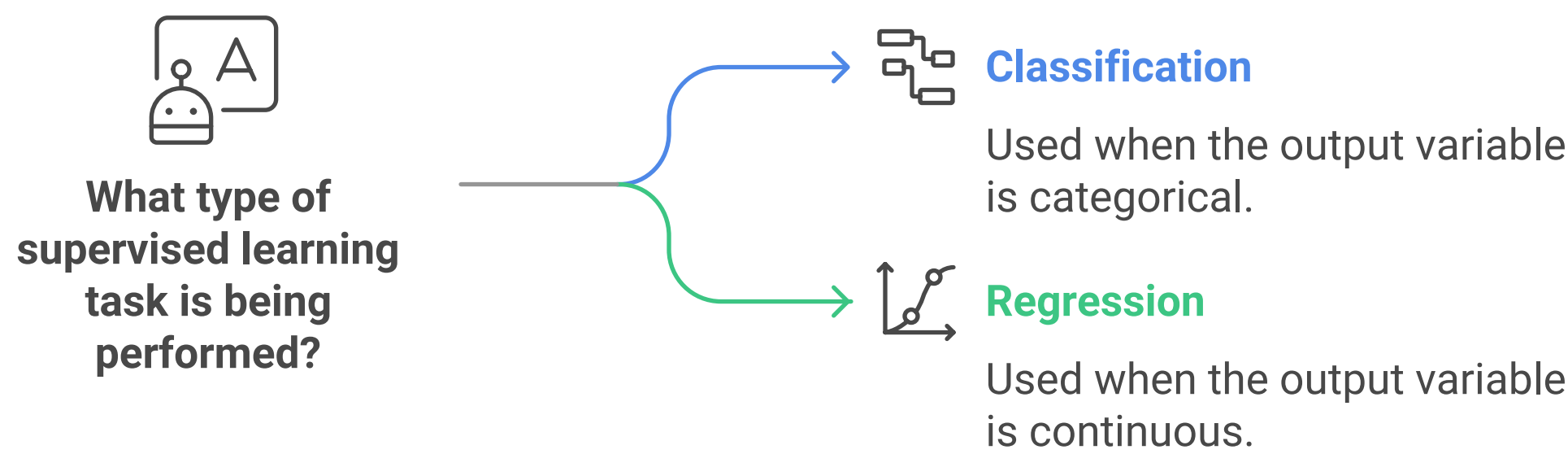
Machine Learning (ML) is broadly categorized into three types: **Supervised Learning**, **Unsupervised Learning**, and **Reinforcement Learning**. Each type has its own unique characteristics, applications, and algorithms.



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Core Learning Paradigms in Machine Learning

1. Supervised Learning (Labeled Data) → Classification & Regression

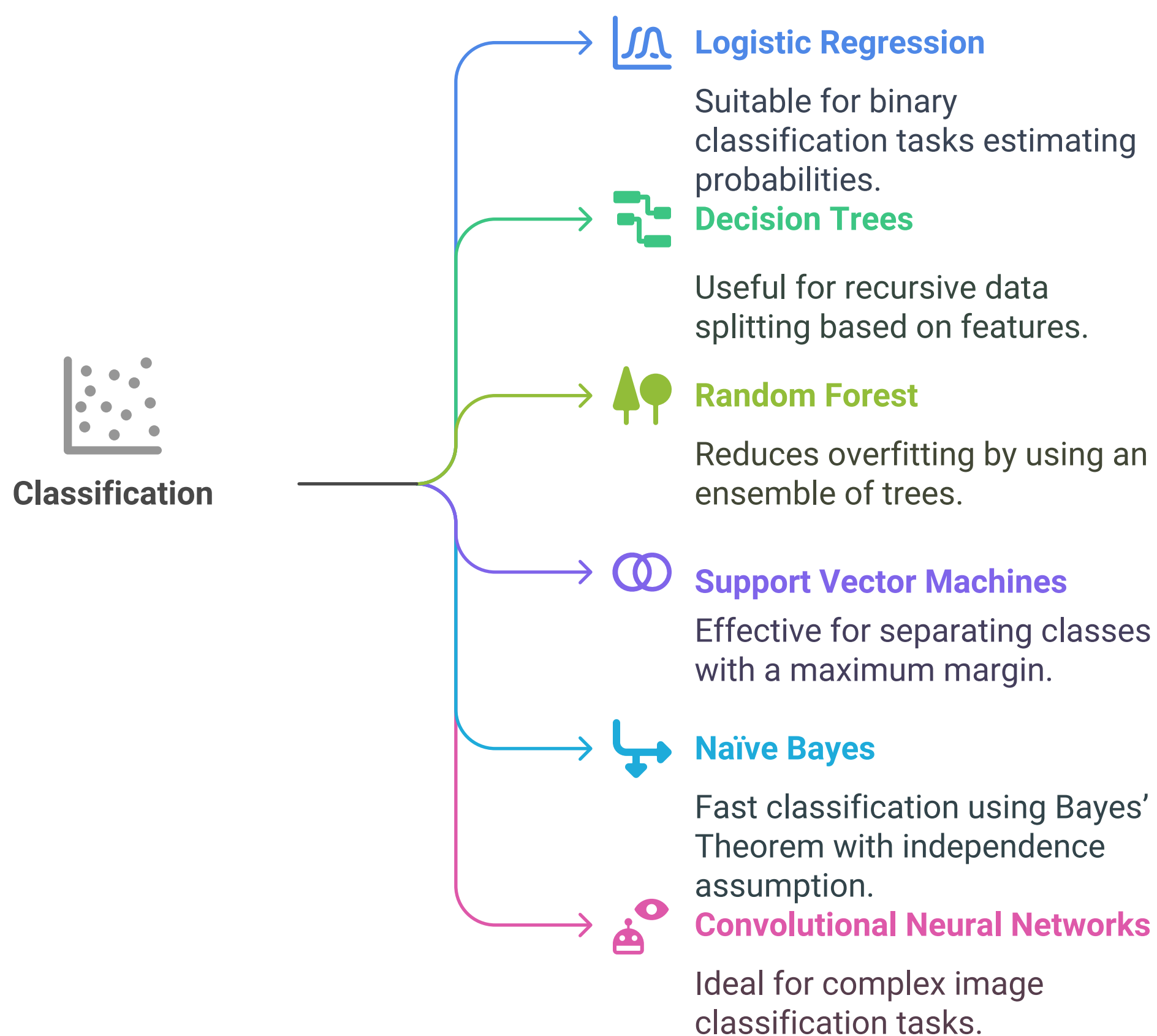


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Supervised Learning involves training a model on labeled data, where each input has a corresponding output.

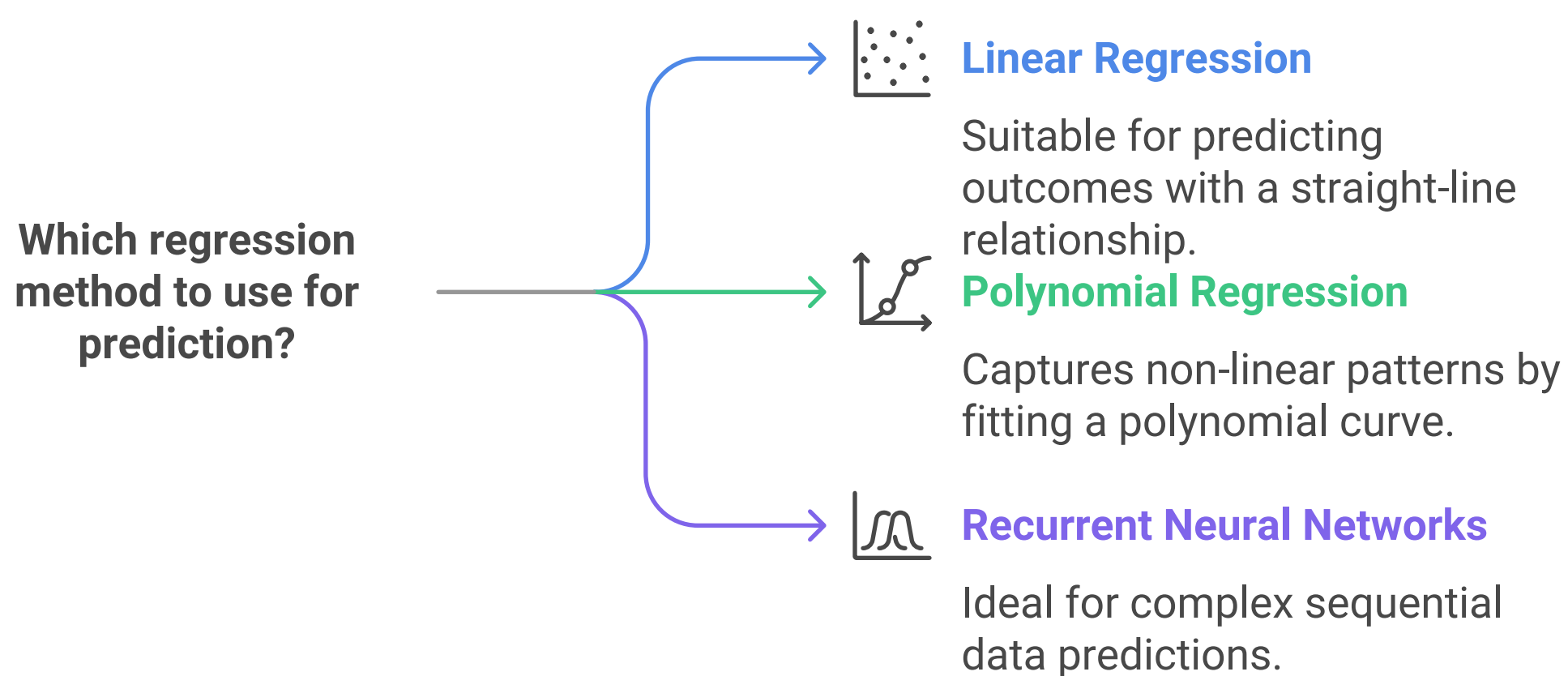
Classification (Predict Discrete Labels)

- **Logistic Regression:** Estimates probabilities using a sigmoid function for binary classification. *Example:* Predicting whether an email is spam or not.
- **Decision Trees:** Splits data recursively based on feature thresholds to classify samples. *Example:* Classifying loan applicants as low or high risk.
- **Random Forest:** An ensemble of decision trees that reduces overfitting and improves accuracy. *Example:* Diagnosing diseases from patient symptoms.
- **Support Vector Machines (SVM):** Finds the optimal hyperplane to separate classes with maximum margin. *Example:* Handwritten digit recognition.
- **Naïve Bayes:** Uses Bayes' Theorem with feature independence assumption for fast classification. *Example:* Sentiment analysis in text reviews.
- **Deep Learning Example** → Convolutional Neural Networks (CNNs): Extracts features from images for complex classification tasks. *Example:* Face recognition in security systems.



Regression (Predict Continuous Values)

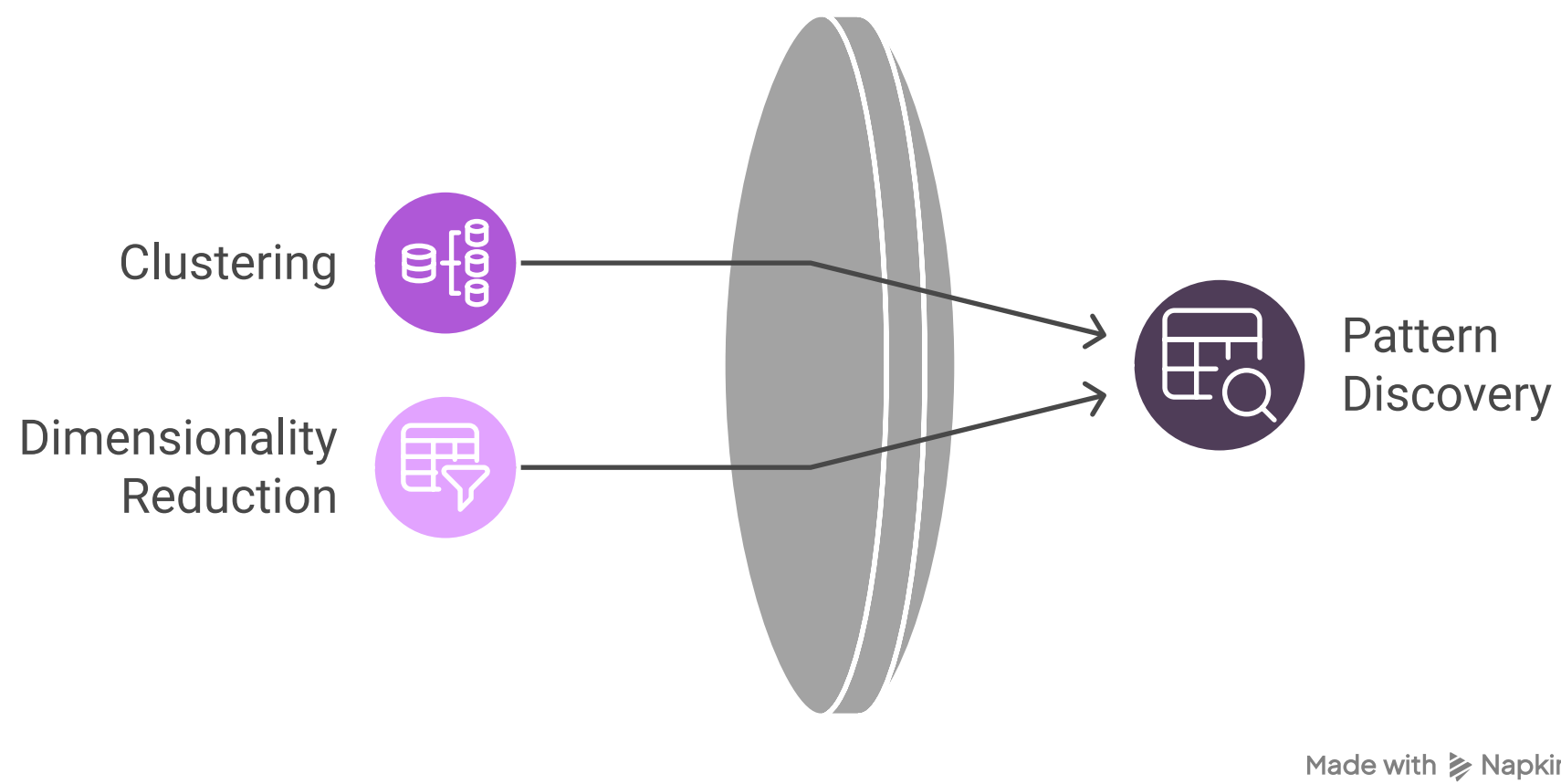
- **Linear Regression:** Models the relationship between input and output using a straight-line equation. *Example:* Predicting house prices based on area.
- **Polynomial Regression:** Extends linear regression by fitting a polynomial curve to capture non-linearity. *Example:* Predicting temperature variations over time.
- **Deep Learning Example** → Recurrent Neural Networks (RNNs): Captures sequential dependencies for complex regression problems. *Example:* Stock price prediction based on historical trends.



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2. Unsupervised Learning (Unlabeled Data) → Clustering & Dimensionality Reduction

Unsupervised Learning Goals

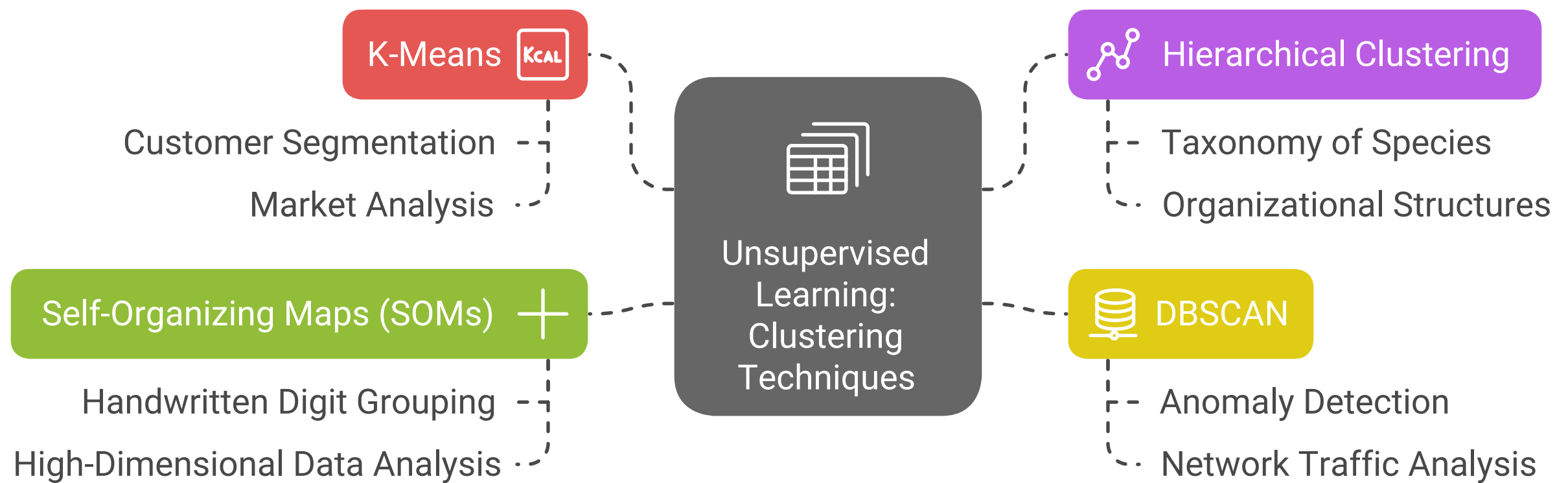


Unsupervised Learning works with unlabeled data to identify hidden patterns and structures.

Clustering

- **K-Means:** Partitions data into K clusters by minimizing intra-cluster variance. *Example:* Customer segmentation in e-commerce.
- **Hierarchical Clustering:** Builds a tree-like structure of nested clusters for flexible grouping. *Example:* Taxonomy of animal species.
- **DBSCAN:** Groups data based on density, identifying clusters of varying shapes. *Example:* Detecting anomalies in network traffic.
- **Deep Learning Example** → Self-Organizing Maps (SOMs): Uses neural networks to discover patterns in high-dimensional data. *Example:* Grouping similar handwritten digits without labels.

Unsupervised Learning: Clustering Techniques and Applications

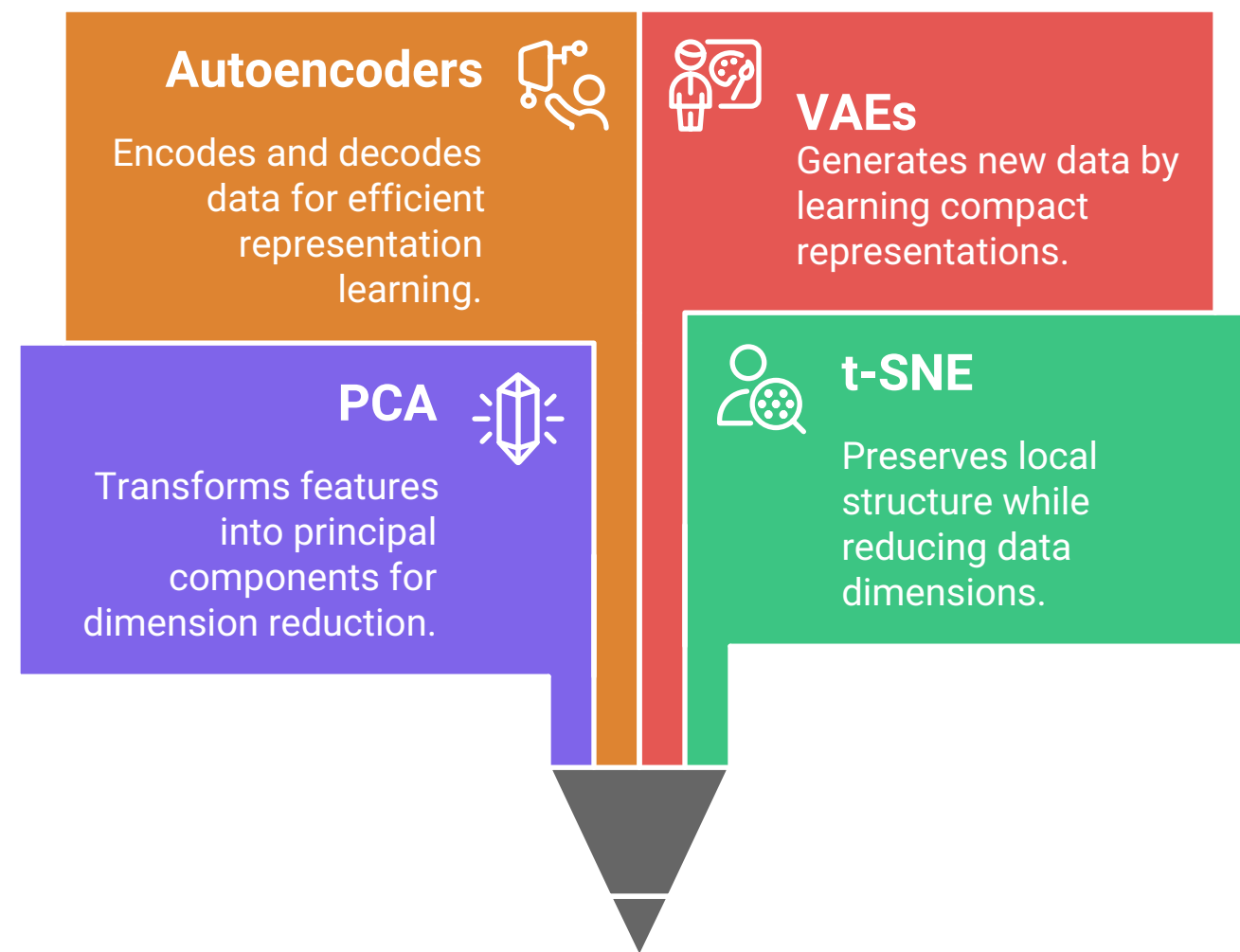


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Dimensionality Reduction

- **Principal Component Analysis (PCA):** Reduces data dimensions by transforming features into principal components. *Example:* Image compression.
- **t-SNE:** Projects high-dimensional data into a lower space while preserving local structure. *Example:* Visualizing word embeddings.
- **Autoencoders:** Neural networks that learn efficient data representations through encoding-decoding. *Example:* Noise reduction in images.
- **Deep Learning Example** → Variational Autoencoders (VAEs): Learns compact representations for complex data distributions. *Example:* Generating new human face images from latent space.

Unified Dimensionality Reduction



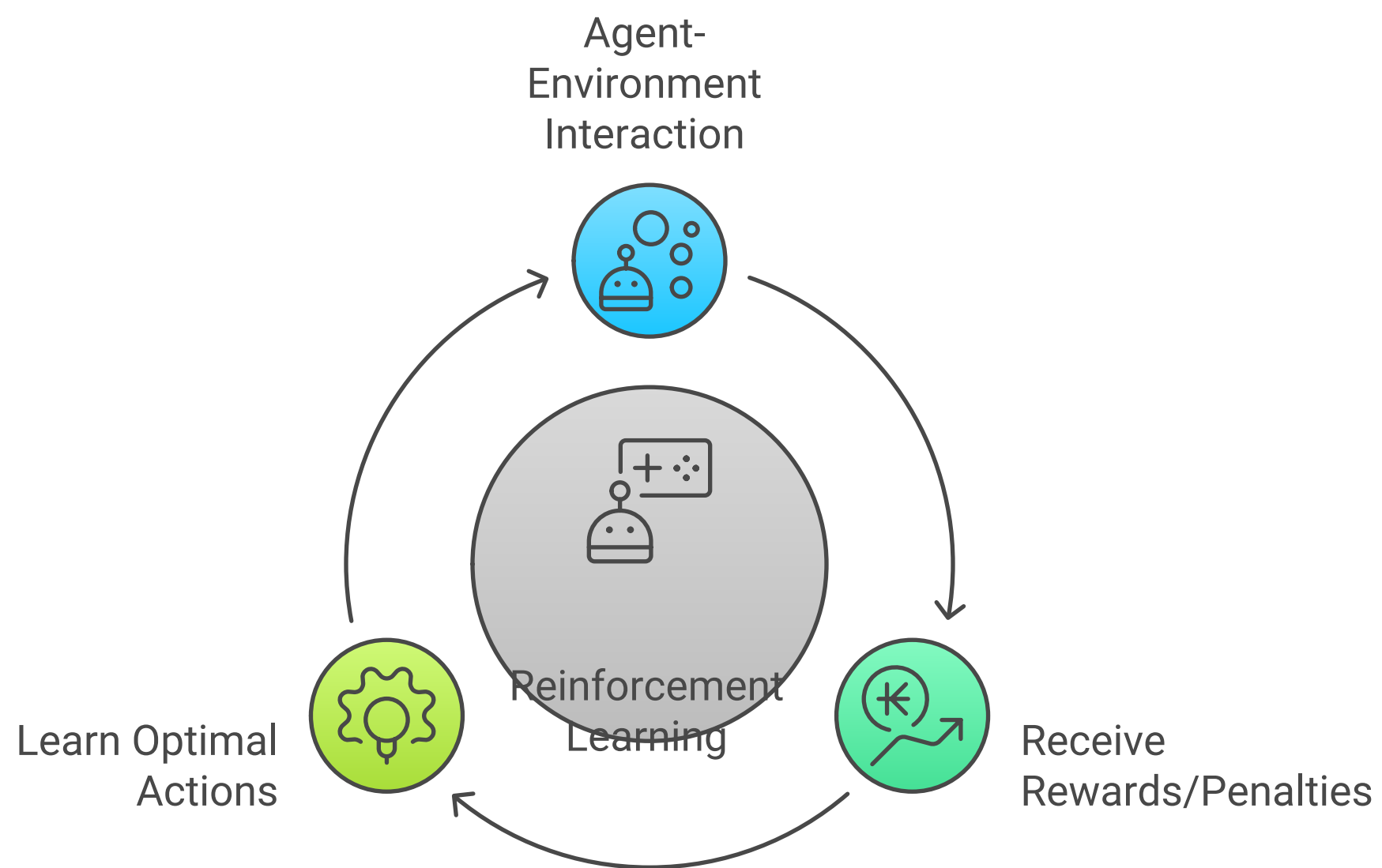
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3.

Reinforcement Learning (Agent-Environment Interaction) → Policy Optimization & Value-Based Learning

Reinforcement Learning trains an agent to interact with an environment and learn optimal actions through rewards and penalties.

Reinforcement Learning Cycle



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Policy Optimization [Directly learn the best action policy]

- **Policy Gradient:** Optimizes a policy function using gradient ascent. *Example:* Controlling robotic arms.
- **Actor-Critic:** Combines policy-based and value-based methods for stable learning. *Example:* Self-driving car decision-making.

- **Deep Learning Example** → Proximal Policy Optimization (PPO): Enhances policy gradient methods for better exploration.*Example:* Training virtual humanoids to walk in simulations.

Which Reinforcement Learning Method to Use?



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Value-Based Learning [Estimates future rewards to make decisions]

- **Q-Learning:** Uses a Q-table to estimate the best action for each state.*Example:* Robot navigation in a maze.
- **Deep Q-Networks (DQN):** Uses neural networks to approximate Q-values for large state spaces.*Example:* Playing Atari games.

- **Deep Learning Example** → AlphaZero: Uses deep neural networks and reinforcement learning to master complex games. *Example:* Playing chess at superhuman levels.

Which reinforcement learning method to use?

